

DS 6021: Intro to Predictive Modeling

Lab 2: Linear Regression

September 11, 2026

Due: September 13, 2026 11:59pm

Materials Required

All materials are available in our class github repository:

- penguins.csv – Palmer penguins dataset
- PlanetsData.csv – Orbital distance and period for the planets
- In-class notebooks from 9/7 (intro_to_linear_regression.ipynb) and 9/9 (linear_regression_part2.ipynb)

Activity Overview

In class this week, we predicted body mass from flipper length, looked at the residuals, did a bootstrap simulation to assess the coefficients, and began adding more than one predictor. We will continue these exercises in this lab.

When we use the penguin dataset, drop rows missing any of the following variables, so we are working from the same dataset across all models:

Part 1: Parameter Estimation and Model Fitting (8 pts)

For this part of the activity, we want to predict body mass (body_mass_g) from bill depth (bill_depth_mm).

- [2 pts]** Create a scatterplot of bill_depth_mm vs. body_mass_g with a fitted regression line. Describe the direction and strength of the relationship. Is the direction what you would have expected?
- [3 pts]** Using only numpy and/or pandas — no sklearn, no statsmodels — compute β_0 and β_1 from the ordinary least squares formulas we derived in class. Report the estimated regression equation, and interpret the slope in the units of the data in one sentence.
- [3 pts]** Calculate RSE using the formula we discussed in class (see slide 35). Also calculate R^2 by first calculating the correlation between bill depth and body mass. What are each of these numbers telling you?

Part 2: How Good Is This Model? (12 pts)

- [2 pts]** Now use statsmodels to predict body mass from bill depth. For the slope (β_1), note the standard error, t-statistic, and p-value. State the null and alternative hypotheses that this t-test is evaluating, and what you can conclude based on these numbers.
- [3 pts]** Now look specifically at the R^2 for this model and the p-value for the slope. Discuss what each of these numbers is measuring, what they are collectively telling you, and how the information you are seeing from each is consistent.
- [7 pts]** Bootstrap the coefficients as we did in class: resample the rows with replacement 10,000 times, calculate β_0 and β_1 each resample, and report the 2.5th and 97.5th percentiles of each as

well as the corresponding confidence interval. Compare these to the 95% confidence intervals statsmodels reports.

- a. Which gives narrower confidence intervals, statsmodels or the bootstrap? What does that tell you?
- b. What assumption does the formula-based interval make that the bootstrap does not?
- c. Change the random seed (or unset it, so it is different each time) and run the bootstrap 10 times with $B=100$. Repeat this for $B=1000$ and $B=10000$. (Note: $B=10000$ in the original question. It is the number of times you repeat the bootstrap simulation.)
- d. For each of these simulations, look at the 2.5th and 97.5th percentiles and view them in a table. You basically want to compare the 95% confidence intervals for each of the 10 simulations at each value of B . What information does this give you about the value of B in your bootstrap simulation?

Part 3: Residuals (12 pts)

- a. [3 pts] Plot fitted values vs. residuals for your bill depth model you just built. Please also replicate the fitted vs residuals plot we looked at in class for the model predicting mass as a function of flipper length. Compare these plots side by side. What is each of these plots telling you about the respective models?
- b. [2 pts] Then plot the same residuals again, colored by species. Describe what you see in each.
- c. [2 pts] Compute the correlation between bill_depth_mm and body_mass_g
 - a. (a) across all penguins and
 - b. (b) separately within each species.
 - c. Explain what is going on.
- d. [5 pts] Build a linear model using bill depth to predict body mass using only data for the Adelie penguins. Create a plot of bill depth vs body mass as well as fitted vs residuals for this model. What conclusions would you draw for just the Adelies? Reflect on how this is related to the conclusions you drew for all penguins.

Part 4: Multiple Linear Regression (8 pts)

For this question we will dive in a bit more deeply into the multiple linear regression models we started looking at during class.

- a. [2 pts] Look at the formulas of R^2 and Adj R^2 that we went over in class. Why do these two values behave differently? How does their behavior differ as more predictors are added. (Hint: The recommended reading will help with insight here as well.)
- b. [3 pts] Fit three models on the same rows, with body_mass_g as the response:
 - a. flipper_length_mm
 - b. flipper_length_mm + bill_depth_mm
 - c. flipper_length_mm + bill_depth_mm + bill_length_mm
 - d. For each of these models, create a table reporting R^2 , adjusted R^2 , RSE, and the F-statistic for each. Which model would you choose, and why?
- c. [2 pts] In Part 1, the simple regression slope on bill depth was negative. Report the coefficient on bill_depth_mm in the second model above. What happened, and why? Use the interpretation of β_j

from class — the average effect of a one unit increase in that predictor, holding the other predictors fixed — in your explanation.

- d. [1 pt] State the null and alternative hypotheses tested by the F-statistic in the third model, and what you conclude from it.

Part 5: Modeling with the Planets Dataset (10 pts)

For this part, switch to PlanetsData.csv, which gives each planet's mean distance from the sun (in millions of miles) and its orbital period, or revolution (in Earth days).

- a. [2 pt] Fit an OLS model predicting revolution from distance and report R^2 . Judging by R^2 alone, how good does this model look? Compare actual and predicted values for each of the planets. Now look at the predicted revolution for the four innermost planets? What does that tell you?
- b. [2 pts] Plot fitted values vs. residuals for this model. What do you see? Which planets have the largest residuals, and in which direction? What does this tell you about the validity of the linear model for this dataset?
- c. [6 pts] One approach that we will discuss further in the coming weeks is transforming variables in order to improve the performance of regression models. We will try that here.
 - a. Apply a log transformation to both distance and revolution and refit the model. This means you will build a linear model with the x variable being $\log(\text{distance})$ and the y variable being $\log(\text{revolution})$.
 - b. Plot $\log(\text{distance})$ vs $\log(\text{revolution})$ as well as fitted vs residuals for this model. Compare this to the model you built in part a. Is this a better fit?
 - c. Report the new R^2 and the estimated slope. The slope should land very close to a simple fraction — what is it? Using this simple fraction and algebra on your fitted equation, show that this implies Kepler's third law of planetary motion: **revolution² $\propto$ distance³**
 - d. With this new equation, create a table of actual vs fitted values for revolution for each planet. How close is the new model to reality vs the linear model?

Deliverables

For each activity, please include the following:

- All code used to complete the activities. This can be turned in as a single Jupyter notebook, multiple Jupyter notebooks, or one or more .py files. Up to you!
- For open ended questions, please include a few sentences or a paragraph answering the question. This text can be included in your Jupyter notebook as a markdown cell or a separate document.
- Per our policy on Generative AI detailed in the syllabus, if you use it to generate code, you need to flag how and where you used it, and how you ascertained the correctness of what you did.

Note: You are not allowed to use Generative AI for any of the reflection questions.

- *If you did not use Generative AI for any part of the assignment, please state that explicitly. (This is new from last week!)*